SEMINAR REPORT

on

Seminar Title

Submitted by

Author

Submitted in partial fulfillment of the requirements for the award of

BACHELOR OF TECHNOLOGY
in
ELECTRONICS AND COMMUNICATION

DIVISION OF ELECTRONICS ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI - 682022

July 2026

DIVISION OF ELECTRONICS ENGINEERING
SCHOOL OF ENGINEERING
COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY
KOCHI-682022

CERTIFICATE

*Certified that the seminar report entitled **"Seminar Title"** is a bonafide work of **Author** towards the partial fulfillment for the award of the degree of B.Tech in Electronics and Communication of Cochin University of Science and Technology, Kochi-682022.*

Staff-in-charge **Head of the Division**

ACKNOWLEDGMENT

Ack here. Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

ABSTRACT

Abstract Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut postea variari voluptas distinguique possit, augeri amplificarique non possit. At etiam Athenis, ut e patre audiebam facete et urbane Stoicos irridente, statua est in quo a nobis philosophia defensa et collaudata est, cum id, quod maxime placeat, facere possimus, omnis voluptas assumenda est, omnis dolor repellendus. Temporibus autem quibusdam et aut officiis debitis aut rerum necessitatibus saepe eveniet, ut et voluptates repudiandae sint et molestiae non recusandae. Itaque earum rerum.

Contents

List of Figures

List of Tables

List of Abbreviations

ASIC:	Application Specific Integrated Circuit
CRT:	Chinese Remainder Theorem
DFT:	Discrete Fourier Transform
DIF:	Decimation in Frequency
DIT:	Decimation in Time
DSP:	Digital Signal Processing
FFT:	Fast Fourier Transform
FT:	Fourier Transform
WNS:	Worst Negative Slack

List of Symbols

- $\int$ - integral
- $\sum$ - sum
- $\partial d/\partial x$ - differential
- $\mathbb{C}, \mathbb{N}$ - complex, natural numbers
- $\mathcal{H}$ - Hilbert
- ∇ - gradient

CHAPTER 1
INTRODUCTION

1.1 Sub section

goto `settings.typ` to change how things are displayed. The file also contains report title. Change `report-title` variable in `settings.typ` to change title of the report. This will be reflected in title page, certificate as well as in headers.

The table of contents, and all sorts of lists are in `tocloft.typ`.

This secton covers some basic constructs. usually there is no header in chapter, reference, and appendix start pages. This is default in many typesetting world

Best practice is to split all chapters into separate `.typ` files, and then include it in `report.typ` main file. This improves maintainability.

CHAPTER 2
FORMATTING

Typst follows markdown like syntax

Bold, *italic*, <u>underline</u> ***bold italic***

inline `monospace`

```
// block code
#include <iostream>
int main () {
  return 0;
}
```

custom text size (not recommended)

to write hash: #

To reference a source: [1]

To reference a section: *Section 1: Introduction*

The formatting for how refs are shown in controlled in `settings.typ`

Use `#pagebreak()` to start new page

2.0.1 lists

- Unnumbered
- Unnumbered
- Unnumbered
- Unnumbered

1. Numbered
    - Un Numbered
2. Numbered
3. Numbered
4. Numbered

2.0.2 Table

header	Header2	Header3
item1	item2	item3
item1	item2	item3
item1	item2	item3

to span whole width of page (use `fr` units):

header	**Header2**	**Header3**
item1	item2	item3
item1	item2	item3
item1	item2	item3

2.0.3 Numbered table

to change caption position see the end of `settings.typ`

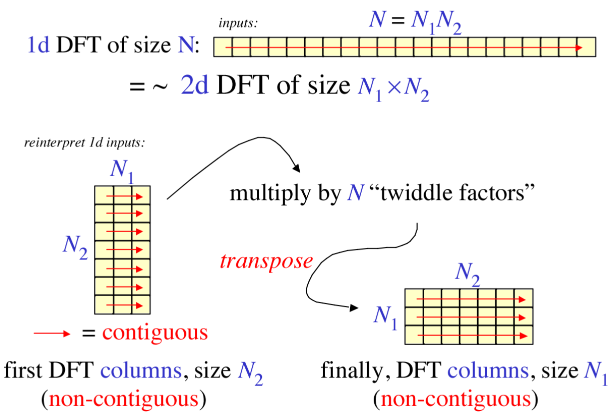
Table 2.1: Table with different stroke style

header	**Header2**	**Header3**
item1	item2	item3
item1	item2	item3
item1	item2	item3

This time a reference was added to table (see code above) to call it: Table 2.1

2.0.4 Figure

Figure 2.1: img caption

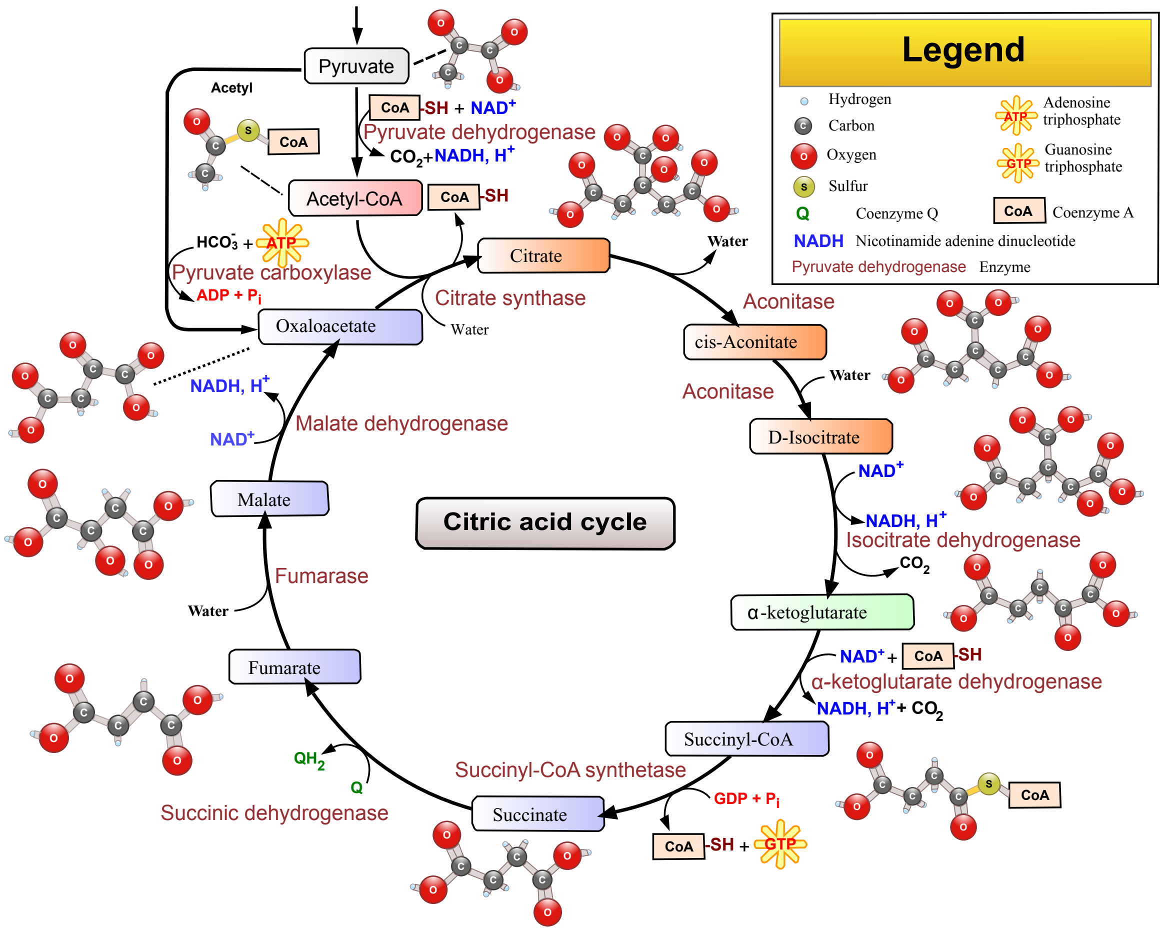


You can also `cetz` (https://typst.app/universe/package/cetz/) to create diagrams in typst itself. See https://github.com/roopeshor/Typesetting/tree/main/Reports/Internship-3 For examples

Some pages needs to cover maximum area of page, to show everything clearly. Manually putting width or height is just hard. And if the image height is more than the page height, you need manual scaling. For that `fit-to-page` function from `settings.typ` can be used, as shown in next page.

Best practice is to put all images in `images/` folder.

Figure 2.2: An image that maximally covers the page. Use fit-to-page function from settings.typ



2.0.5 column layout

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum no-

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magnam aliquam quaerat voluptatem. Ut enim aeque doleamus animo, cum corpore dolemus, fieri tamen permagna accessio potest, si aliquod aeternum et infinitum impendere malum nobis opinemur. Quod idem licet transferre in voluptatem, ut.

bis opinemur. Quod idem licet transferre in voluptatem, ut.

2.0.6 math

Inline math: $\sum_{n=0}^{\infty} n^{-2}$

block math:

$$\sum_{n=0}^{\infty} n^{-2} = \frac{\pi^2}{6}$$

aligned equations:

$$\begin{aligned}
\frac{-b \pm \sqrt{b^2 - 4ac}}{2a} &= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\
&= -\frac{b}{2a} \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\
&= -\frac{b}{2a} \pm \sqrt{\frac{b^2}{4a^2} - \frac{4ac}{4a^2}} \\
&= -\frac{b}{2a} \pm \sqrt{\frac{b^2}{4a^2} - \frac{c}{a}}
\end{aligned}$$

$$\int_{x=0}^{\infty} \frac{d}{dx} x^2 dx$$

if multiple letters are there, do seperate them by space, or typst will treat it as some variable.

2.0.6.1. matix:

$$= \begin{bmatrix} 1 & 2 & 4 \\ 1 & 3 & 4 \\ 5 & 6 & 7 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$$

2.0.6.2. numbered equation

$$X(k) = \sum_{n=0}^{5} \exp\left(-\frac{2\pi}{5} nk\right) = 0 \tag{2.1}$$

To ref it: Equation (2.1) or to get just number: (2.1). Formatting can be changed in `settings.typ`

REFERENCES

[1] Xilinx, "7 Series FPGAs Data Sheet: Overview." [Online]. Available: https://docs.amd.com/v/u/en-US/ds180_7Series_Overview

[2] Yongchul Jung, Jaechan Cho, Seongjoo Lee, and Yunho Jung, "Area-Efficient Pipelined FFT Processor for Zero-Padded Signals." [Online]. Available: https://www.mdpi.com/2079-9292/8/12/1397

[3] P. Duhamel and M. Vetterli, *Fast Fourier Transforms: A Tutorial Review and a State of the Art*. [Online]. Available: https://dsp-book.narod.ru/DSPMW/07.PDF

[4] I.J. Good, "The Relationship Between Two Fast Fourier Transforms." [Online]. Available: https://ieeexplore.ieee.org/document/1671829

[5] Nikhilesh Bhagat, Daniel Valencia, Amirhossein Alimohammad, and Fred Harris, "High-Throughput and Compact FFT Architectures Using the Good-Thomas and Winograd Algorithms." [Online]. Available: https://amir.sdsu.edu/Bhagat18High.pdf

APPENDICES

A.1 Source Code

For more examples. `https://github.com/roopeshor/Typesetting`

A.2 Section 2

sdfs